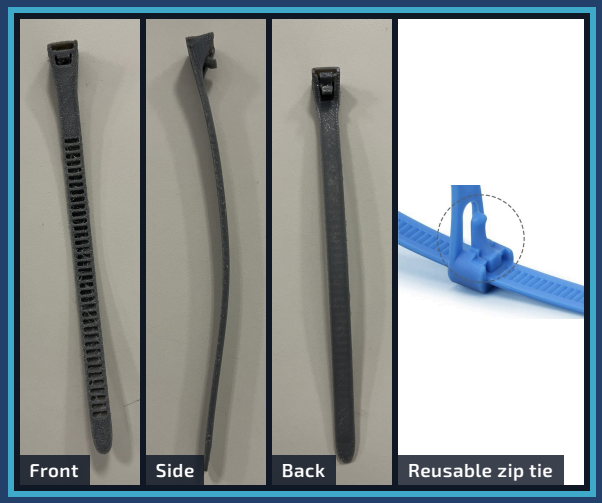
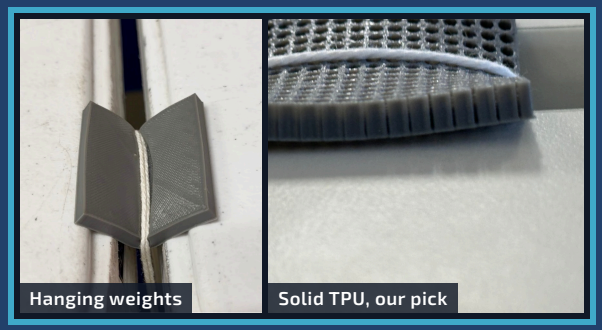
Our prototype: (Straps)

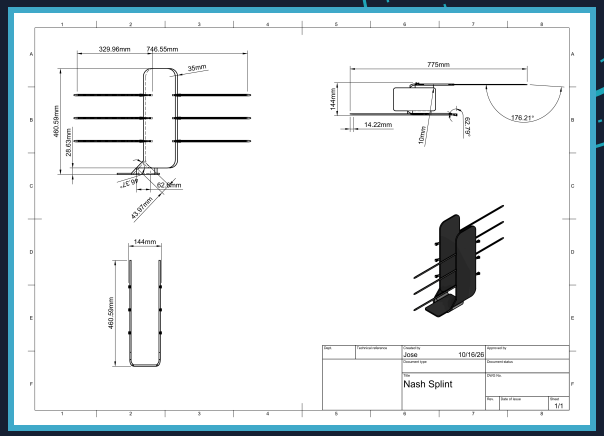


(Testing)

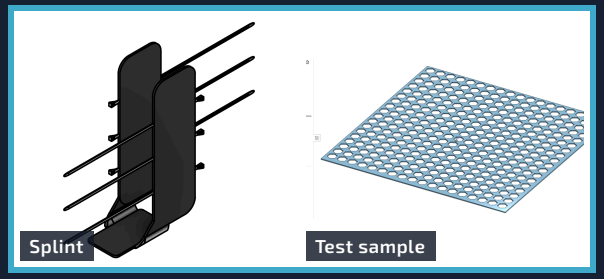


- Custom fit from a scan
- One-handed straps
- Printed on the Moon

(Detail drawing)



(CAD)



WE FOCUS ON FIT, FUNCTION, AND THE MOON.

A student design project. Not a medical device.

ORTHOPRINT



NASA HUNCH
Lunar Splint Design Challenge

Gladys Porter Early College High School
Brownsville, Texas

Team Lunova:
Gertrudis, Jose, & Nancy

Teachers:
Lupita C. Vasquez, Claudia Cortez

WHY ORTHOPRINT

- A custom fit from a scan of the patient's own leg
- Straps the astronaut can tighten with one hand
- Printed on the base, so no box of spare sizes to launch
- Tested in PLA, ABS, TPU and PETG

VISION & MISSION

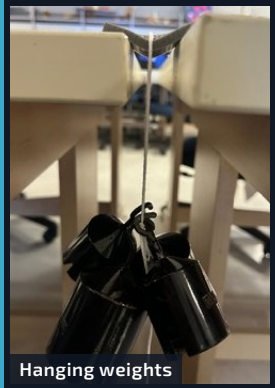
- Give an injured astronaut a splint made for their own ankle
- Hold a torn ankle ligament still so it can heal
- Print what the crew needs, when they need it
- Stay clean and comfortable over weeks of wear

(Samples)



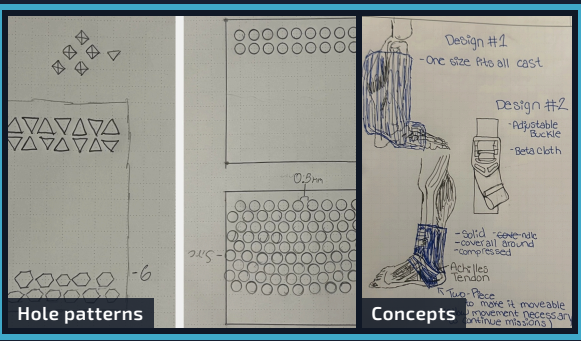
Holes split

(Load test)



Hanging weights

(Design sketches)



Hole patterns

Concepts

Website:

D.A.A.N.:

WHO WE ARE

We are Team Lunova, three students who designed OrthoPrint: a splint for a torn ankle ligament that a lunar base can 3D-print from a scan of the astronaut's leg.



TPU sample

WHAT WE OFFER

- **Single-piece ratchet closure**
  Tightens or releases with one hand, no loose parts.
- **Integrated hygiene**
  Comes off to clean and check the skin.
- **Thermal phase design**
  Flexible to put on, firm to hold.